La Yaung Phyo

Bangkok, Thailand | +66 98 385 6507 | layaungphyo079@gmail.com
linkedin.com/in/layaungphyo | github.com/FreddieTheObserver | layaungphyo.dev

Education

King Mongkut's University of Technology Thonburi (KMUTT) Bangkok, Thailand
BSc in Computer Science, School of Information Technology. GPA 3.47 / 4.00 *2025 - 2029 (expected)*

- Coursework: Data Structures, Discrete Mathematics, Database Systems, Computer Architecture and Organization, System Analysis and Design, Linear Algebra for Computer Science Applications

Projects

floodwatch | *Go, PostgreSQL, Telegram Bot API, Docker, GitHub Actions* Sep 2026

- Built and launched a Thai and English Telegram bot that warns Bangkok residents when nearby canals approach their banks or rain turns heavy, live on the day the city declared all 50 districts flood disaster areas.
- Polls public water level and rain gauge feeds every 10 minutes into PostgreSQL, keying readings by observation time and recording an alert only after Telegram accepts it, so retries never duplicate or lose one.
- Added a tide model that forecast each tidal gauge's 6-hour peak within 8 cm on average in a backtest, against 45 cm for a no-change baseline.

CineBook | *Go, PostgreSQL, sqlc, Docker, GitHub Actions* Sep 2026

- Built a cinema booking API in which a seat can never be sold twice: per-showtime PostgreSQL advisory locks serialize every write, with a partial unique index behind them as a hard invariant.
- Designed a two-phase flow of expiring holds and idempotent confirms, proven by a contention test, run under the Go race detector in CI, in which 200 goroutines race for one seat and exactly one wins.

CafePOS | *TypeScript, NestJS, PostgreSQL, Redis, Stripe, WebSockets* Jul 2026 - Nov 2026

- Built the backend of a cafe point-of-sale system in which self-order kiosks, a counter POS and a real-time kitchen display share one order pipeline, with server-side pricing and idempotent order requests.
- Integrated Stripe so that payment state follows the signed gateway webhook and never the client, with refunds, reconciliation and expiry of unpaid orders.
- Wrote 43 integration suites that run against real PostgreSQL, Redis and S3-compatible storage in CI, including Redis outage and crash recovery cases.

Neural Language Model Paper Reproduction | *Python, PyTorch* Jul 2026

- Reimplemented the MLP language model of Bengio et al. (2003) on the Brown corpus. It lowered test perplexity by 24.1% against a Kneser-Ney trigram baseline, matching the paper's roughly 24%, and by 29.5% when mixed with the trigram.
- Traced a result that contradicted the paper, where four words of context lost to two, to overfitting and fixed it with weight decay.

CrisisTrade, SIT Hackathon 2026 winner | *TypeScript, React, Express, Prisma* May 2026

- Won the hackathon in a team of four with a disaster marketplace where people trade supplies, donate, post urgent requests and find safe exchange points when shops and money stop working.
- Owned the Marketplace Trades feature end to end, from the REST API and data model to the React UI: trade posts filtered by area, urgency, item and status, with reputation points awarded when a trade completes.

Experience

CS Mentor, CSC102 Introduction to Programming Aug 2026 - Present
School of Information Technology, KMUTT *Bangkok, Thailand*

- Guide first-year students through Java fundamentals on site and online, and write short practice sets on arrays, strings and object-oriented design.

Open Source Contributor Sep 2026
unishare-oss/unishare, a student platform for sharing notes and past papers *TypeScript*

- Found and documented two setup steps missing from the contributor guide (the pgvector requirement and a code generation step) in a merged pull request.

Technical Skills

Languages: TypeScript, Java, Python, SQL, Go
Backend and Data: PostgreSQL, Redis, NestJS, Express, REST APIs, WebSockets, Stripe, sqlc, Drizzle, Prisma
Frontend and ML: React, Tailwind CSS, PyTorch, NumPy
Tools: Docker, Linux, Git, GitHub Actions, testcontainers
Spoken Languages: Burmese (native), English (IELTS 7.5)